

Continuing understanding of proof of Theorem 2.1

- Recap:

(proof by induction)

- Base case: h π -monotone embedding

if h is not connected (Multiple components)

we consider each subgraph induced by a comp. of h as our graph after proving by induction that $\forall$ Comp has a π -monotone emb. (with π pos) (same pos)

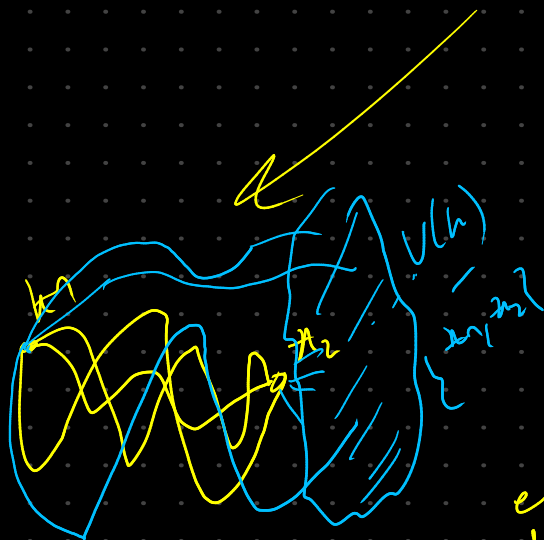
we concat vertically all comps $\rightarrow h$ has π -monotone embedding.

$\rightarrow$ induction?
consider

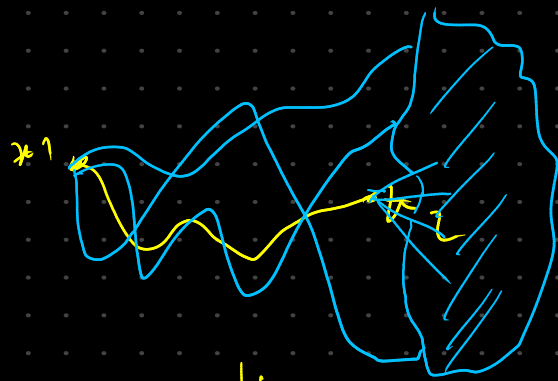
we have h

$x_1, x_2 \in V(h)$ with $x_1 < x_2$

$\forall x \in V(h) - \{x_1, x_2\} \quad x > x_2$



$\Downarrow$ $(e_1, \dots, e_k)$ edges from x_1 to x_2



$\Downarrow$

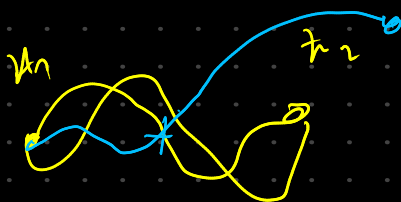
Not x_2 on the left
have consecutive

$(e_1 - e_2)$

[there are the only
edges on the left
of x_2]

for $x_1, \dots, x_n$ on the
right

Suppose



must cross
a $e_i, i \in [1, k]$
oddly. $\Downarrow$

conley $(e_1 - e_k)$
in Rot_{x_1}



move x_2 along
the edge (x_1, x_2)

every cross on
 (x_1, x_2)

that was removed
after moving x_2
creates a cross
with each edge
with x_2 as left
endpoint.



even cross of (x_1, x_2)

and other edges

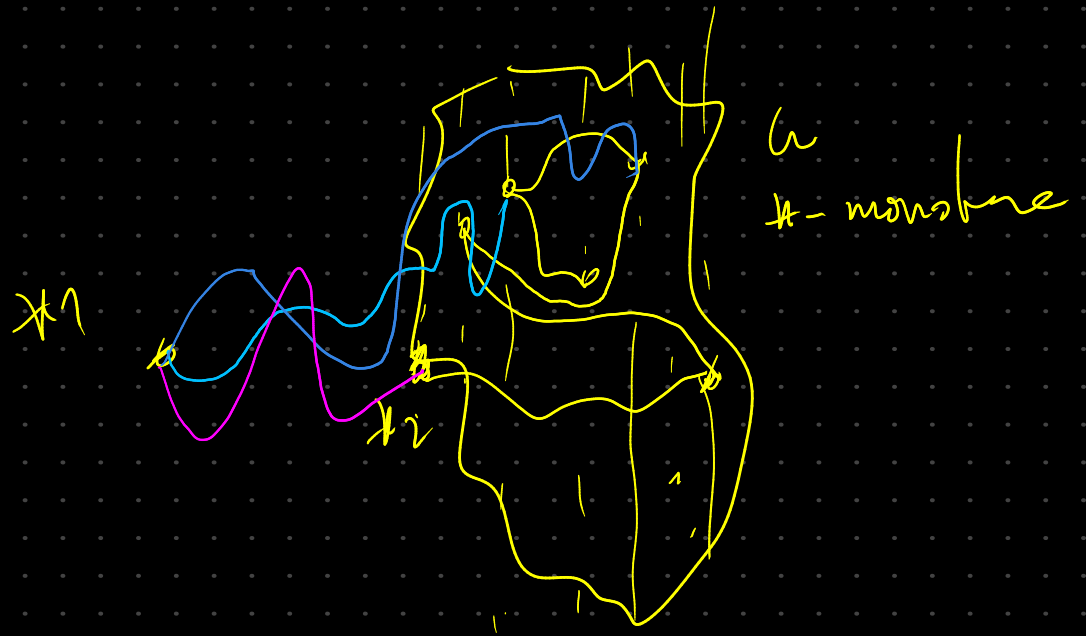


remain with even
cross after merging
 x_2 with x_1 .

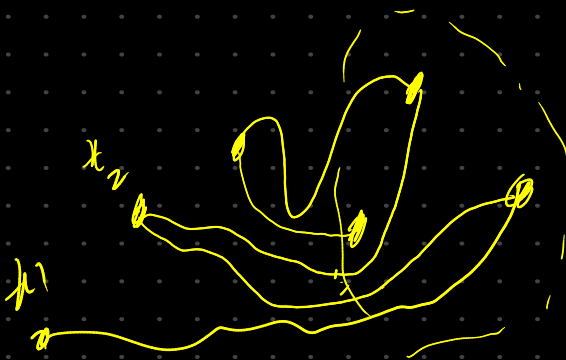
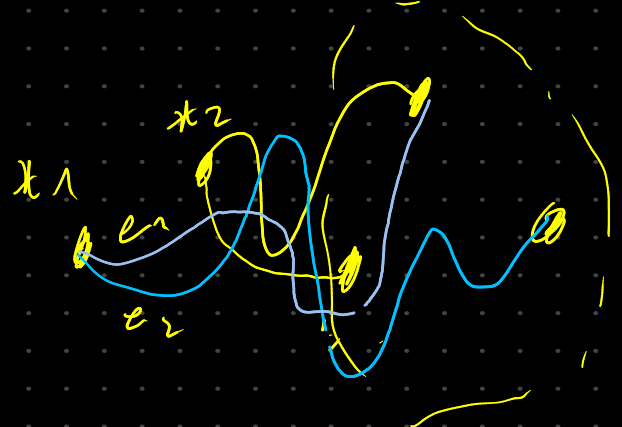


we have $\mathbb{A}$ -monotone
embedding with

$x_1 = x_2 < x_3 < \dots < x_n$



$e \rightarrow f$ (rotation)
 $f \in e$ in $\mathbb{R}^n$

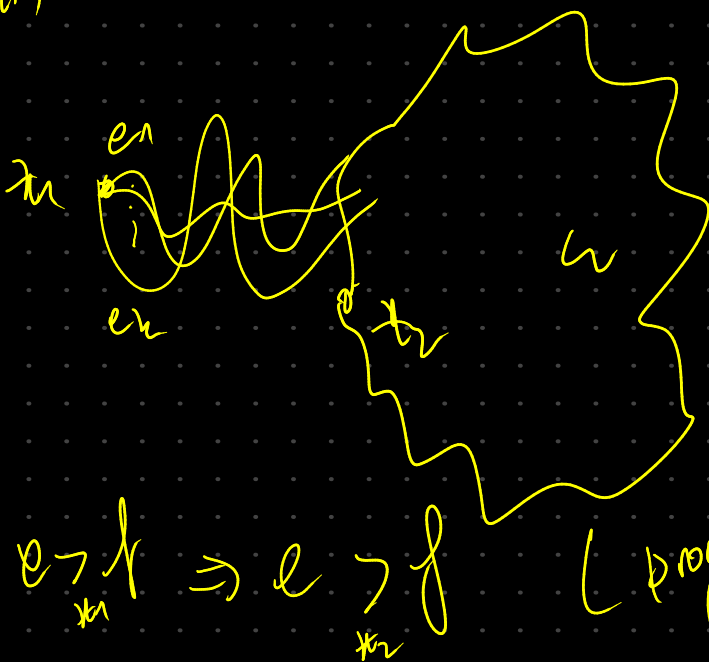


if we translate
 x_2 to x_1 ,
 we are
 in a graph
 with same vertices
 as V_{H_1} adding
 more edges
 that are even
 to a level
 planar graph



→ You can find more
 on this in
 proof - Her - 2.1 - x_1 - x_2 -
 one edge: 1 pe

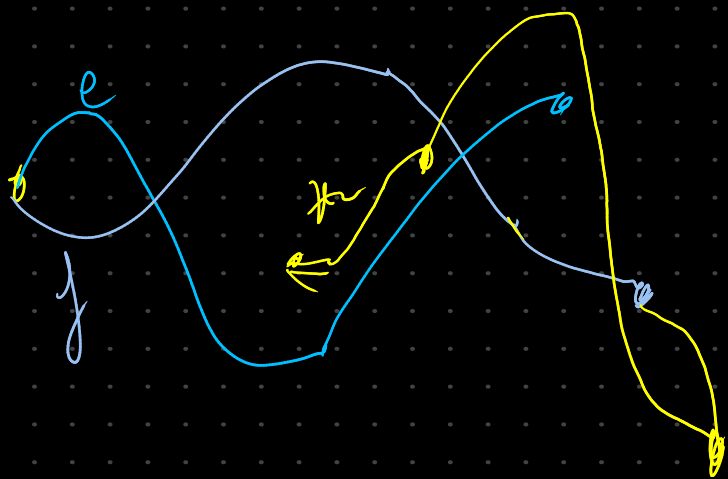
* Cases depend on each other.
 (cont)



with $h - \{x_1\}$
 single comp.

distinct e_i above x_2
 below x_2

$e \rightarrow f \Rightarrow e \rightarrow f$ (proof by contradiction)



a cycle $\mathcal{C}$
 $e, f \in \mathcal{C}$
 $\mathcal{C}$ is connected
 so adding e, f
 to w should
 lead to
 a cycle

$$e = (x_i, x_j)$$

$$f = (x_i, x_j)$$

$P(x_i, \dots, x_j)$ already exists in w (is connected)
 new path (x_i, x_i, x_j) with $e, f \Rightarrow$ cycle $\mathcal{C}$

$w - x_i$ s. comp

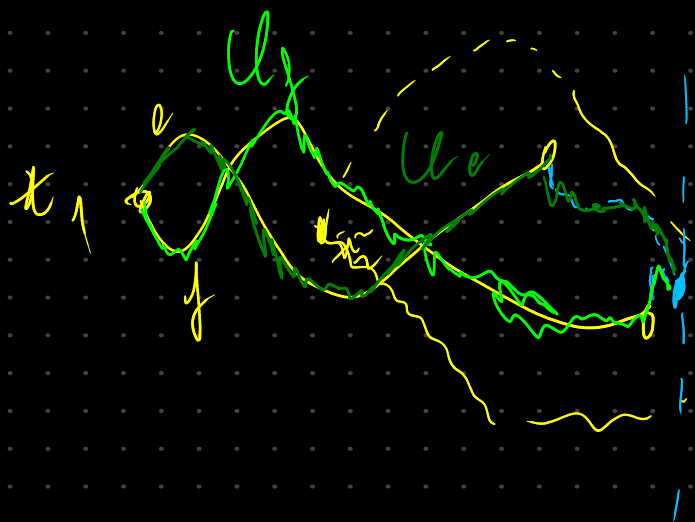
let $\Pi_{\mathcal{C}}$ be the rightmost vertex of $\mathcal{C}$.

- curve C_e starts below x_2 on e and leads to $\Pi_{\mathcal{C}}$.
- curve C_f starts just above x_2 on f leads to $\Pi_{\mathcal{C}}$.

shortest path: P_{x_2} from x_2 to $\mathcal{C}$ ($\mathcal{C}$ is connected so it exists)

(i) P_{x_2} strictly to the left $\Pi_{\mathcal{C}}$.

(ii) P_{x_2} contains a vertex at n at the right of $x = \Pi_{\mathcal{C}}$



for (i) without being gen.
suppose P_{x_2} ends on ℓ_f . (it is connected to a vertex on ℓ_f)

P'_{x_2} is π_2, π_1 -subpath of $P_{x_2} \cup \ell_f$.

since ℓ_e and P'_{x_2} share no edges.

and ℓ_e below x_2 it must be also below

π_1 since there is P'_{x_2} from x_2 to π_1
and edges are even (contradiction)
 ℓ_f must be below ℓ_e
so all edges are even
but $\ell_e \not\geq \ell_f$

ii) P_{x_2} contains a vertex at or to the right of $x = \pi_1$.

Let P'_{x_2} be s. subpath of P_{x_2} ($x_2 \rightarrow \pi_1$)

↓
this means P_{x_2} has no common edge with ℓ_f and

ℓ_e (has to find a directed path since it went till π_1 or even on the right of it)

if it prints ℓ_e it will continue in it till π_1
and go back from ℓ_f not via v_1, v_2 .

$\Rightarrow P'_{x_2}$ enters M_C above ℓ_e or below ℓ_f
 if P'_{x_2} ends in M_C $\ell_e > \ell_f$
 gap!

otherwise: P'_{x_2} passes M_C above ℓ_e and below ℓ_f



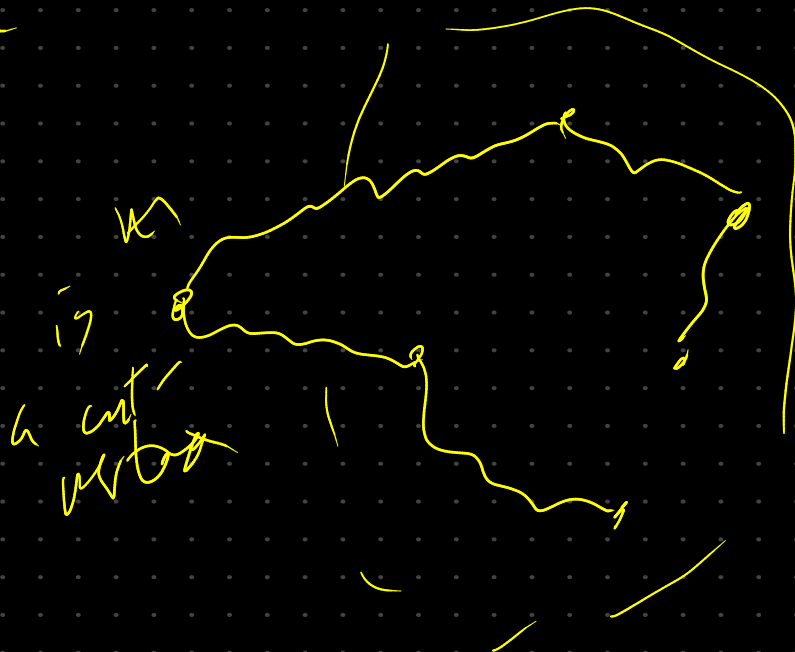
that contradict

$$\ell_f > P'_{x_2}$$

$$\left\{ \begin{array}{l} \ell_f < P_{x_2} \\ \ell_e > P'_{x_2} \end{array} \right. \text{ but}$$

$$\Rightarrow \ell_e < P_{x_2}$$

(contradiction) [odd crossing]



What if v_1 is a cut-vertex

(G is not necessarily connected,
but we can have an embedding
by merging component vertically together.)